

Technology Stack

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	<i>HTML, CSS, Bootstrap</i>	<i>Creates a responsive and user-friendly interface for displaying dashboards and project information.</i>

Technology Stack Details

2	Backend / Server-Side	<i>Python, Flask</i>	<i>Handles application logic, database connectivity, and serves web pages efficiently.</i>
3	Database / Data Storage	<i>Microsoft SQL Server</i>	<i>Stores electricity consumption data securely and supports efficient SQL queries for analysis.</i>
4	Cloud / Hosting / Deployment	<i>Tableau Public</i>	<i>Publishes interactive dashboards online for easy access and visualization.</i>

5	Version Control & CI/CD	<i>Git, GitHub</i>	<i>Tracks project changes, enables collaboration, and maintains version history.</i>
6	Third-Party APIs / Other Tools	<i>Tableau Public, Microsoft SQL Server, Visual Studio Code</i>	<i>Used for interactive visualization, database management, and application development.</i>